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Wernersson et al.

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(54) **ASYMMETRIC VERTICAL NANOWIRE MOSFET HAVING ASYMMETRIC NANOWIRE GEOMETRY NEAR METAL GATE AND METHOD OF FABRICATING THEREOF**

(58) **Field of Classification Search**

CPC H01L 21/823412; H01L 21/823418; H01L 29/66742; H01L 29/78618; H01L 29/78696; H01L 21/823487

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 641 days.

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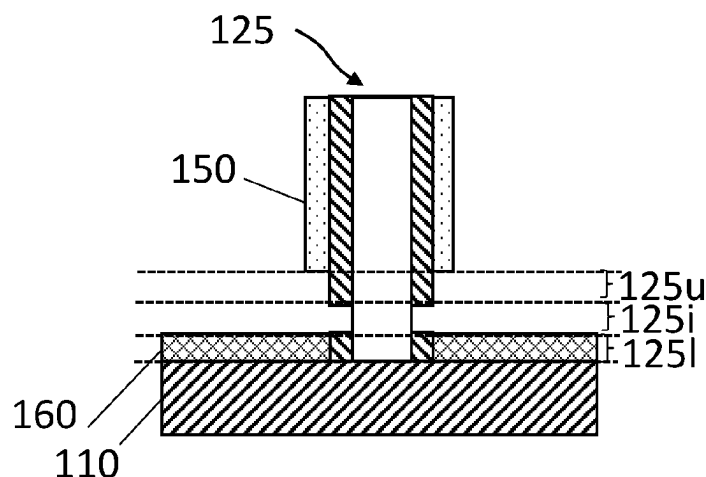
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ABSTRACT

There is provided a method for fabricating an asymmetric vertical nanowire MOSFET on a semiconductor substrate comprising at least one vertical nanowire, comprising a core portion and a shell portion circumscribing the core portion. The method comprises depositing a protection layer on the semiconductor substrate, forming a top contact around a remaining portion of the vertical nanowire not covered by the protection layer, removing the protection layer, depositing a spacer layer on the semiconductor substrate, removing a shell portion of the intermediate portion of the bottom portion of the vertical nanowire, trimming a shell portion of

(Continued)



the upper portion of the bottom portion of the vertical nanowire, depositing a metal gate on the spacer layer, and forming a lower and an upper source drain portions.

8 Claims, 4 Drawing Sheets

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H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
- (52) **U.S. Cl.**
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- (58) **Field of Classification Search**
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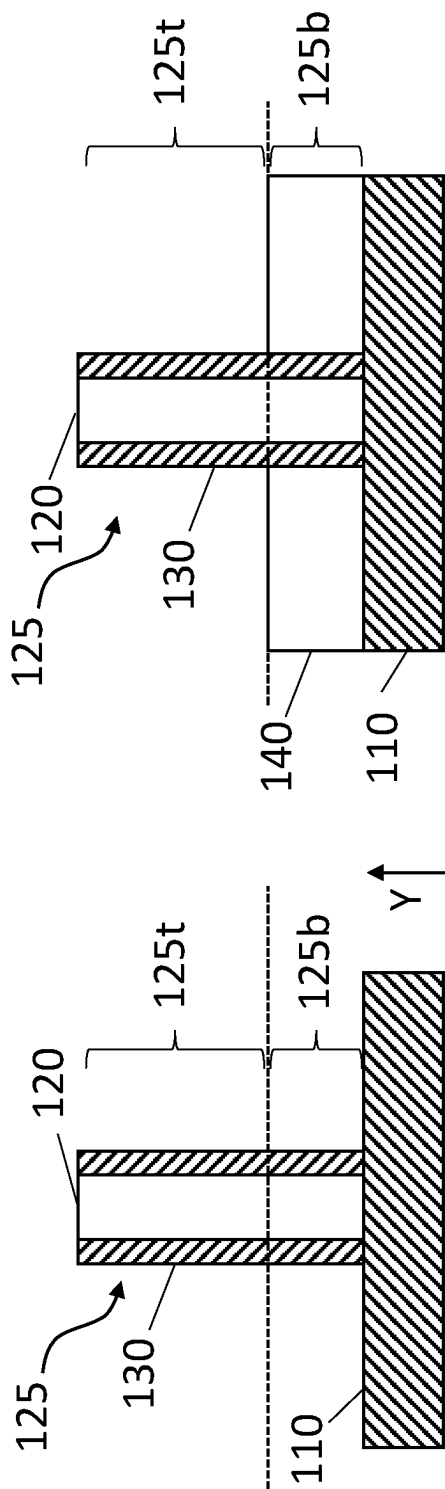


Fig. 1

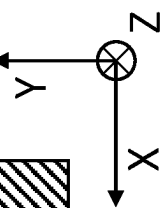


Fig. 2

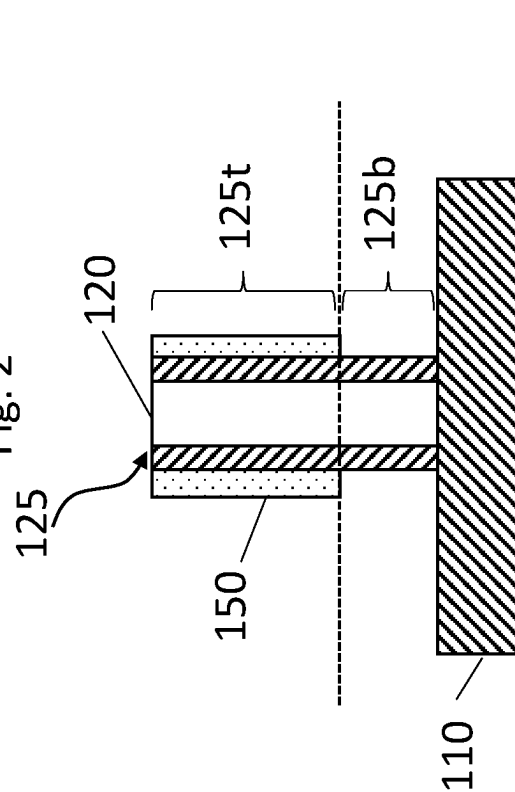


Fig. 2

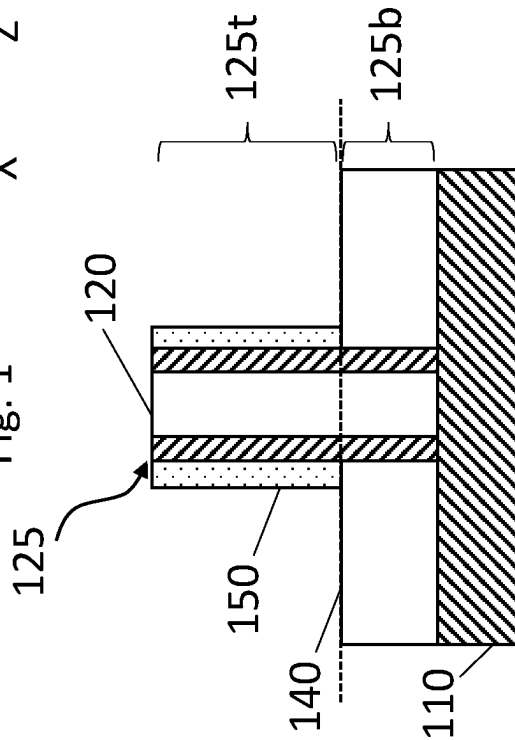


Fig. 3

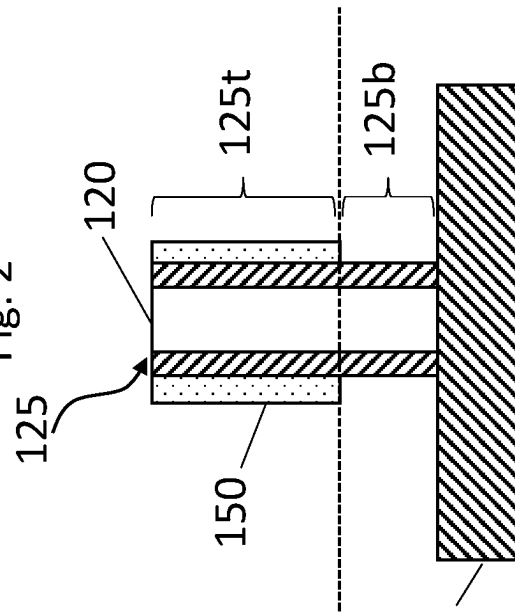


Fig. 4

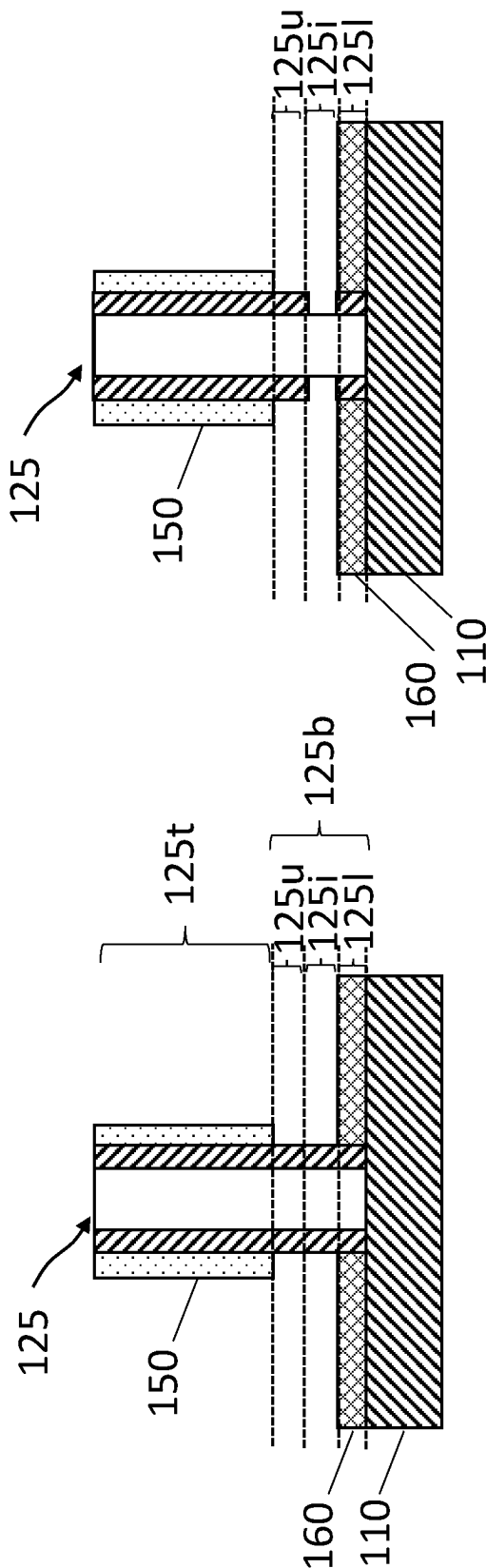


Fig. 6

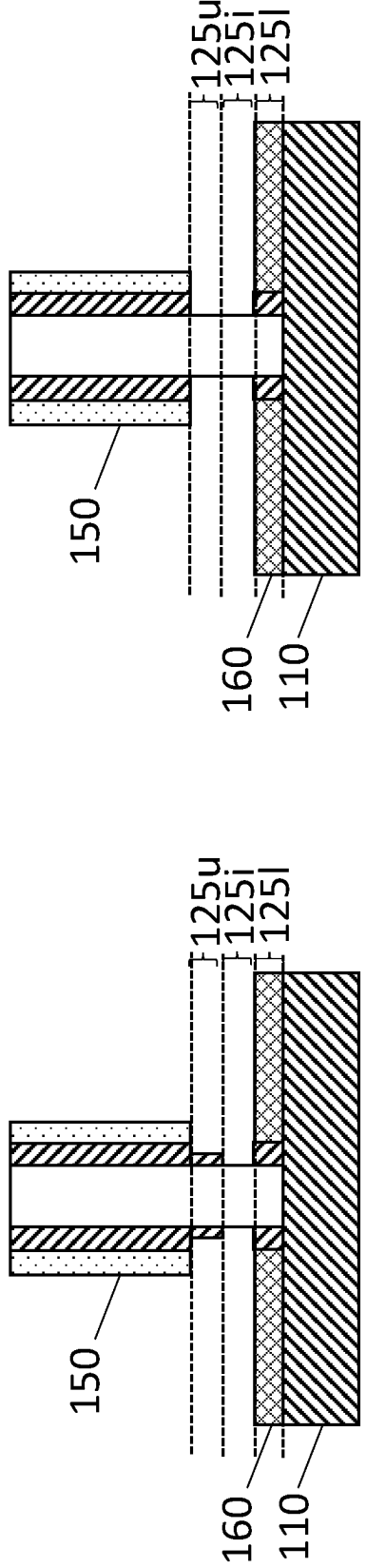


Fig. 7b

Fig. 5

Fig. 7a

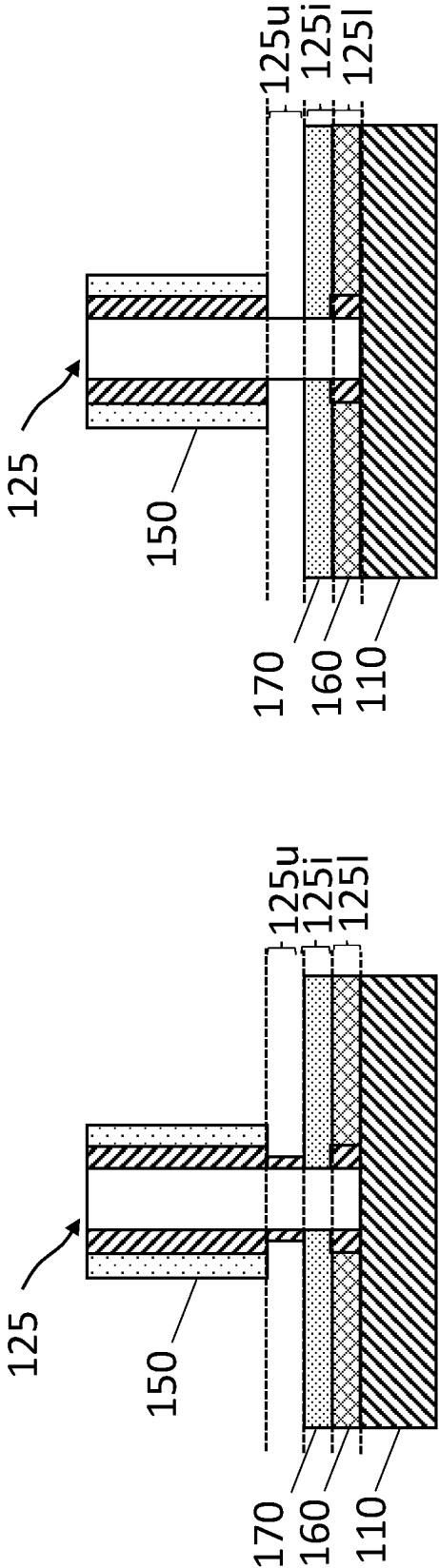


Fig. 8a

Fig. 8b

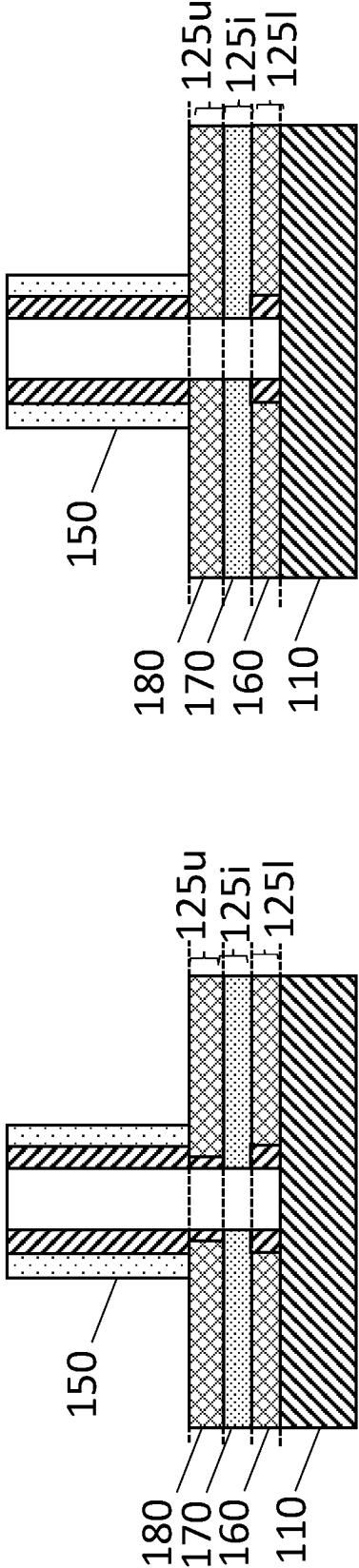


Fig. 9a

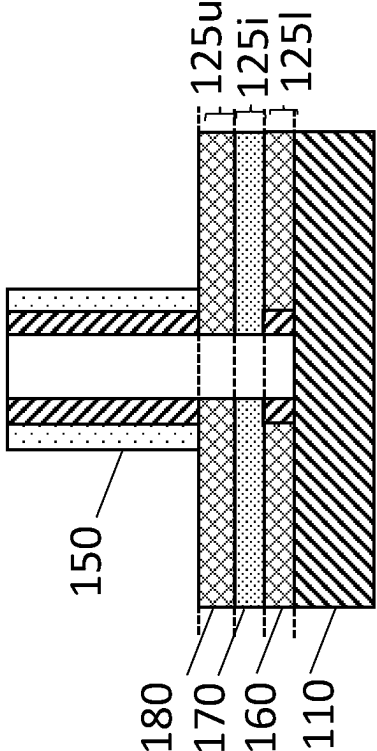


Fig. 9b

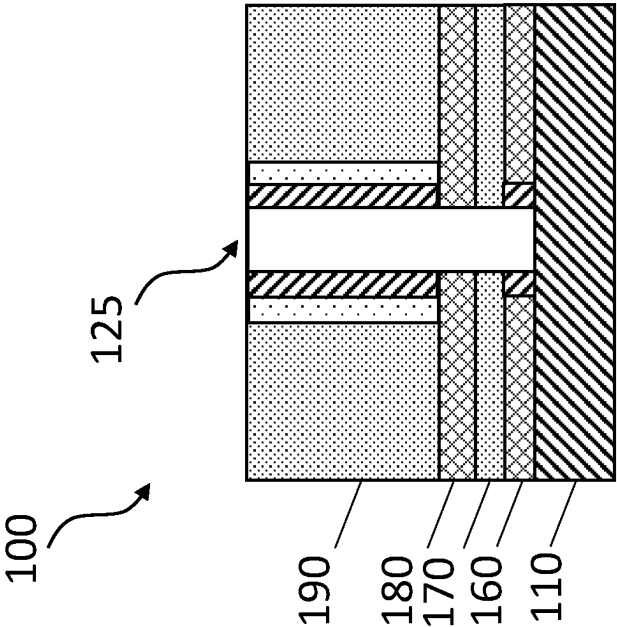


Fig. 10b

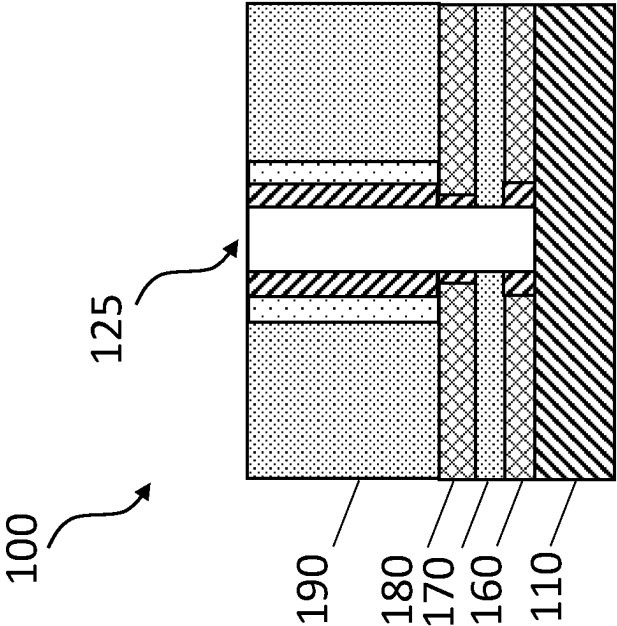


Fig. 10a

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ASYMMETRIC VERTICAL NANOWIRE MOSFET HAVING ASYMMETRIC NANOWIRE GEOMETRY NEAR METAL GATE AND METHOD OF FABRICATING THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This Application is the National Stage filing under 35 U.S.C. § 371 of PCT Application Ser. No. PCT/EP2020/075437 filed on Sep. 11, 2020, which claims the benefit of Swedish Patent Application No. 1951042-9 filed on Sep. 16, 2019. The disclosures of both applications are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present inventive concept relates to a method for fabricating an asymmetric vertical nanowire metal oxide semiconductor field effect transistor (MOSFET) on a semiconductor substrate and an asymmetric vertical nanowire MOSFET.

BACKGROUND

Over the past years, vertical nanowire MOSFET devices have been developed allowing gate-all-around geometry. In particular, MOSFETs formed of III-V materials integrated on Si substrates may offer a possibility to decrease the supply voltage and hence to decrease a power consumption. However, vertical nanowire MOSFETs have shown rather low performance, compared to the state-of-art lateral MOSFETs. The lower performance of the vertical nanowire MOSFETs may be related to their longer gate length, higher access resistances and higher capacitances, compared to the lateral MOSFETs.

SUMMARY

An object of the present inventive concept is to, at least partly, overcome one or more of the above-identified limitations of the prior art.

According to an aspect of the present inventive concept there is provided a method for fabricating an asymmetric vertical nanowire MOSFET on a semiconductor substrate comprising at least one vertical nanowire, wherein the at least one vertical nanowire comprises a core portion and a shell portion circumscribing the core portion along a height of the nanowire, the method comprising: depositing a protection layer on the semiconductor substrate and around a bottom portion of the vertical nanowire, forming a top contact around a remaining portion of the vertical nanowire not covered by the protection layer, removing, by etching, the protection layer, depositing a spacer layer on the semiconductor substrate and around a lower portion of the bottom portion of the vertical nanowire, wherein a thickness of the spacer layer is smaller than the thickness of the protection layer and wherein a remaining portion of the bottom portion of the vertical nanowire, not covered by the spacer layer, comprises an intermediate portion and an upper portion, removing a shell portion of the intermediate portion of the bottom portion of the vertical nanowire, thereby exposing a core portion of the vertical nanowire at the intermediate portion of the bottom portion of the vertical nanowire, trimming a shell portion of the upper portion of the bottom portion of the vertical nanowire, thereby reduc-

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ing a thickness of the shell portion of the upper portion of the bottom portion of the vertical nanowire or removing the shell portion of the upper portion of the bottom portion of the vertical nanowire, depositing a metal gate on the spacer layer and around the intermediate portion of the bottom portion of the nanowire, and forming a lower and an upper source drain portions.

The at least one vertical nanowire comprising a core portion and a shell portion allows e.g. a combination of two different materials. The step of depositing a protection layer on the semiconductor substrate and around a bottom portion of the vertical nanowire allows for protecting the bottom portion of the vertical nanowire during the step of forming the top contact. Thereby, the top contact is formed on the remaining portion of the vertical nanowire i.e. a top portion of the vertical nanowire not covered by the protection layer. In addition, a thickness of the protection layer may e.g. define a spacing between the top contact and the lower source drain portion. The step of removing, by etching, the protection layer allows access to the bottom portion of the vertical nanowire while the top portion of the nanowire is covered by the top contact. The step of depositing the spacer layer on the semiconductor substrate and around a lower portion of the bottom portion of the vertical nanowire allows defining the spacing between the gate and the lower source drain portion. For instance, a thickness of the spacer layer may affect the spacing between the gate and the lower source drain portion.

The steps of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire allow forming an asymmetric portion of the vertical nanowire arranged in close proximity of the metal gate i.e. the lower portion of the bottom portion of the vertical nanowire and the upper portion of bottom portion of the vertical nanowire become asymmetric. For instance, a thickness of the upper portion of the bottom portion of the vertical nanowire becomes smaller than a thickness of the lower portion of bottom portion of the vertical nanowire. The asymmetric vertical nanowire MOSFET in turn allows engineering a gate-source capacitance (C_{gs}) and a gate-drain capacitances (C_{gd}) such that asymmetric C_{gs} and C_{gd} may be achieved. At the same time, the presence of the shell on in the direct vicinity of the metal gate, at least on one side of the metal gate, provides a mean to decrease the access resistance that is crucial for digital and radio frequency (RF) applications. The asymmetric C_{gs} and C_{gd} may in turn improve a power gain of an obtained asymmetric vertical nanowire MOSFET.

By a vertical nanowire MOSFET is hereby meant a MOSFET comprising a semiconductor structure comprising a lower and an upper source drain portion and a channel portion located intermediate and extending vertically between the lower and upper source drain portions, and further comprising a gate structure extending vertically along the channel portion. The gate structure may at least partially enclose the channel portion. In particular, the gate structure may wrap-around the channel portion, in other words forming a gate-all-around (GAA) structure. The lower and upper source drain portions and the channel portion may intersect a common vertical plane. The channel portion is adapted to (in use of the device) conduct a vertical flow of charge carriers between the source drains.

As used herein, the term “vertical” denotes a direction or orientation (e.g. of a surface, a dimension or other feature) parallel to a normal direction of the substrate (i.e. a main plane of extension or main/upper surface thereof). The term

“horizontal” meanwhile denotes a direction or orientation parallel to the substrate (i.e. a main plane of extension or main surface thereof), or equivalently transverse to the vertical direction.

For instance, by a “vertical nanowire” is hereby meant that the nanowire height extends parallel to a normal of the semiconductor substrate.

By “on” is hereby meant above and in contact with. For instance, by “depositing a protection layer on the semiconductor substrate” is hereby meant that the protection layer is deposited above and in contact with the semiconductor substrate.

Meanwhile, terms such as “bottom”, “top”, “lower” and “intermediate”, “upper” refer to relative positions as viewed along the vertical direction and does hence not imply an absolute orientation of the substrate or device. For instance, the terms “bottom portion” and “top portion” of the nanowire refer to relative positions viewed along the vertical direction i.e. the “bottom portion” is arranged closer to the semiconductor substrate than the “top portion”. In addition, the terms “lower”, “intermediate”, and “upper” portions of the bottom portion of the vertical nanowire refer to relative positions viewed along the vertical direction i.e. the “lower portion” is arranged closer to the semiconductor substrate than the “intermediate portion” and the “upper portion” and the “intermediate portion” is arranged closer to the semiconductor substrate than the “upper portion”. The terms “lower source drain portion” and “upper source drain portion” refer to relative positions of the source drain portions viewed along the vertical direction.

By “asymmetric vertical nanowire MOSFET” is hereby meant that the nanowire geometry is not symmetric in close proximity of the metal gate i.e. the lower portion and the upper portion of the bottom portion of the vertical nanowire are asymmetric.

The step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire may comprise removing the shell portion of the upper portion of the bottom portion of the vertical nanowire, thereby exposing a core portion of the vertical nanowire at the upper portion of the bottom portion of the vertical nanowire. Thereby an asymmetric vertical nanowire MOSFET may be achieved i.e. the lower portion of the bottom portion of the vertical nanowire may have a shell portion while the upper portion of the bottom portion of the vertical nanowire may have no or a little shell portion.

The step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire may be performed by etching. Etching may facilitate removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire. For instance, the etching may be performed such that a rate of etching may be controlled. This may in turn allow an improved control during the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire.

The step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire may be performed simultaneously. Thereby, the fabrication of an asymmetric vertical nanowire MOSFET may be facilitated. For

instance, the number of processing steps to fabricate an asymmetric vertical nanowire MOSFET may be reduced. By “simultaneously” is hereby meant that the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire are done in a same process and at the same time.

The method may further comprise masking the upper portion of the bottom portion of the vertical nanowire, prior to the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire. Thereby, the upper portion of the bottom portion of the vertical nanowire may be protected during the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire. This may in turn provide an improved control on the fabrication of the asymmetric vertical nanowire MOSFET.

The step of depositing the metal gate on the spacer layer and around the intermediate portion of the bottom portion of the vertical nanowire may be performed prior to the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire. Thereby the metal gate may protect the intermediate portion of the bottom portion of the vertical nanowire during the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire. In other words, the metal gate may mask the intermediate portion of the bottom portion of the vertical nanowire during the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire.

The method may further comprise unmasking the upper portion of the bottom portion of the vertical nanowire, prior to the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire. The unmasking may thereby allow accessing the upper portion of the bottom portion of the vertical nanowire prior to the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire.

The core portion of the vertical nanowire may be formed of InAs, InGaAs or a combination thereof. The core portion of the vertical nanowire may be formed of a binary material, a ternary material or a combination thereof. The binary and the ternary materials may be formed of any of III-V semiconductor materials. The core portion of the vertical nanowire may be doped with an n-type or a p-type dopant.

The shell portion of the vertical nanowire may be formed of InGaAs with an n-type dopant dose of at least 1.10^{16} atoms/cm³. The shell portion of the vertical nanowire may be formed of a binary material, a ternary material or a combination thereof. The binary and the ternary materials may be formed of any of III-V semiconductor materials. The shell portion of the vertical nanowire may be doped with an n-type or a p-type dopant. A dose of the dopant may be in a range of 1.10^{16} to 1.10^{20} atoms/cm³.

According to another aspect of the present inventive concept there is provided an asymmetric vertical nanowire MOSFET comprising: at least one vertical nanowire arranged on a semiconductor substrate, wherein the at least one vertical nanowire comprises a core portion and a shell portion circumscribing the core portion along a height of the vertical nanowire, a first spacer layer arranged on the semiconductor substrate and around a lower portion of a bottom portion of the vertical nanowire, a metal gate arranged on the first spacer layer and around an intermediate portion of the bottom portion of the vertical nanowire, wherein the intermediate portion of the bottom portion of the vertical nanowire comprises a through-going recess in the

shell portion, the recess circumscribing the core portion, a second spacer layer arranged on the metal gate and at least around an upper portion of the bottom portion of the vertical nanowire, wherein the shell portion at the upper portion of the bottom portion of the vertical nanowire has a reduced thickness compared to the lower portion of the bottom portion of the vertical nanowire, a top contact arranged around a top portion of the vertical nanowire, wherein the shell portion of the top portion of the vertical nanowire has a corresponding thickness compared to the lower portion of the bottom portion of the vertical nanowire, and a lower and an upper source drain portions.

This aspect may generally present the same or corresponding advantages as the former aspect.

By "through-going recess" in the shell portion of the intermediate portion of the bottom portion of the vertical nanowire is hereby meant that there is no shell portion or a little shell portion at least at a portion of the intermediate portion of the bottom portion of the vertical nanowire. In other words, the at least a portion of the intermediate portion of the bottom portion of the vertical nanowire is void of any shell portion.

By "top contact" is hereby meant a contact portion arranged at the top portion of the vertical nanowire. The top contact may be a portion of the upper source drain portion. The top contact may be connected to the upper source drain portion.

The asymmetric vertical nanowire MOSFET may comprise a first vertical nanowire having a first diameter and a second vertical nanowire having a second diameter different from the first vertical nanowire. Thereby the asymmetric vertical nanowire MOSFET may operate at different threshold voltages.

The first vertical nanowire may have a first core portion and a first shell portion, and the second vertical nanowire may have a second core portion and a second shell portion. The first and the second vertical nanowires may be formed such that a first core portion may have a same diameter as the second core portion and the first shell portion may have a different diameter than the second shell portion. The first and the second vertical nanowires may be formed such that a first core portion may have a different diameter than the second core portion and the first shell portion may have a same diameter as the second shell portion. The first and the second vertical nanowires may be formed such that a first core portion may have a different diameter than the second core portion and the first shell portion may have a different diameter than the second shell portion.

The asymmetric vertical nanowire MOSFET may further comprise a gate oxide layer arranged in between the core portion of the intermediate portion of the bottom portion of the vertical nanowire and the metal gate. The gate oxide may facilitate modulation of a conductance of the asymmetric vertical nanowire MOSFET channel.

BRIEF DESCRIPTION OF THE DRAWINGS

The above, as well as additional objects, features and advantages of the present inventive concept, will be better understood through the following illustrative and non-limiting detailed description, with reference to the appended drawings. In the drawings like reference numerals will be used for like elements unless stated otherwise.

FIGS. 1-10 schematically illustrate various steps of a method for fabricating an asymmetric vertical nanowire MOSFET on a semiconductor substrate.

DETAILED DESCRIPTION

Various methods for fabricating an asymmetric vertical nanowire MOSFET on a semiconductor substrate will now be disclosed in connection with FIGS. 1-10.

FIG. 1 shows a cross-sectional view of semiconductor substrate 110 comprising at least one vertical nanowire 125. In FIG. 1, the axis Y denotes a vertical direction corresponding to a normal direction with respect to an upper surface of the semiconductor substrate 110. The axis X denotes a first horizontal direction along the semiconductor substrate 100 and the axis Z denotes a second horizontal direction along the semiconductor substrate 100, perpendicular to the first direction X. It should be noted that in the drawings the relative dimensions of the shown elements, such as the height, width or thickness of elements, is merely schematic and may, for the purpose of illustrational clarity, differ from a physical structure.

The semiconductor substrate 110 may be any conventional substrate 110, such as a substrate suitable for MOSFET processing. The substrate 110 may for instance be a semiconductor substrate such as a Si substrate, a Ge substrate or a silicon-germanium (SiGe) substrate. Other examples include a silicon-on-insulator (SOI) substrate, a GeOI substrate or a SiGeOI substrate. The substrate may comprise a stack of layers such as an InAs layer formed on a Si substrate. The InAs layer may be epitaxially grown on the Si substrate. The InAs layer may be epitaxially grown in a manner which per se is known in the art. The InAs layer may be formed using any suitable conventional technique, such as chemical vapor deposition (CVD), metalorganic CVD (MOCVD), molecular beam epitaxy (MBE) or vapor phase epitaxy (VPE). A typical thickness of the InAs layer may be in a range of 100 to 500 nm.

In FIG. 1 and the following figures, one vertical nanowire 125 is shown. However, it should be noted that the semiconductor substrate 110 may comprise any number of further vertical nanowires 125 outside the illustrated region.

The vertical nanowire 125 shown in FIG. 1 extends in parallel to the Y direction. By way of example, a typical height of the vertical nanowire 125 may be in a range of 300 to 600 nm. Shorter heights in the range of 50 to 300 nm may also be considered. A diameter of the vertical nanowire 125 i.e. along the X axis or the Z axis may be in a range from 15 to 100 nm. Diameters of 5 to 15 nm may also be considered. The vertical nanowire 125 may be formed in a manner which per se is known in the art, e.g. by etching an epitaxial semiconductor layer or layer stack formed on the semiconductor substrate 110. In the case of the InAs layer formed on the Si substrate, the vertical nanowire 125 may be formed by etching the InAs layer formed on the Si substrate. The vertical nanowire 125 may be epitaxially grown on the semiconductor substrate 110 in a manner which per se is known in the art. For instance, the vertical nanowire 125 may be grown using catalyst nanoparticles such as Au nanoparticles. The vertical nanowire 125 may be formed using any suitable conventional technique, such as CVD, MOCVD, MBE or VPE.

FIG. 1 shows that the vertical nanowire 125 comprises a core portion 120 and a shell portion 130 circumscribing the core portion 120 along a height of the vertical nanowire. The core portion 120 of the vertical nanowire 125 may be formed of InAs, InGaAs, or a combination thereof. The core portion 120 of the vertical nanowire 125 may comprise a stack of layers along the vertical direction i.e. along the height of the vertical nanowire 125. In other words, the core portion 120 of the vertical nanowire 125 may comprise an axial hetero-

structure. The core portion **120** of the vertical nanowire **125** may comprise a stack of layers perpendicular to the vertical direction i.e. along the width of the vertical nanowire **125**. In other words, the core portion **120** of the vertical nanowire **125** may comprise a radial heterostructure. Any layer of the stack of layers may be formed of a binary material or a ternary material formed of III-V semiconductor materials. The core portion **120** of the vertical nanowire **125** may be doped with an n-type or p-type dopant. A diameter of the core portion **120** of the vertical nanowire **125** i.e. along the X axis or the Z axis may be in a range from 5 to 80 nm.

The shell portion **130** of the vertical nanowire **125** may be formed of InGaAs. The shell portion **130** of the vertical nanowire **125** may comprise a stack of layers. Any layer of the stack of layers may be formed of a binary material or a ternary material formed of III-V semiconductor materials. The shell portion **130** of the vertical nanowire **125** may be doped with an n-type or p-type dopant. In the case of n-type dopant, the n-type dopant dose of the shell portion **130** of the vertical nanowire **125** may be in a range of 1.10^{16} to 1.10^{20} atoms/cm³. The n-type dopant dose of the shell portion **130** of the vertical nanowire **125** may at least be 1.10^{16} atoms/cm³. A diameter of the shell portion **130** of the vertical nanowire **125** i.e. along the X axis or the Z axis may be in a range from 3 to 20 nm.

The core portion **120** and the shell portion **130** of the vertical nanowire **125** may be formed in a manner which per se is known in the art. For instance, in the case of epitaxial growth of the vertical nanowire **125**, the growth of the core portion **120** and the shell portion **130** of the vertical nanowire **125** may be performed by selecting appropriate precursors controlling epitaxial growth parameters e.g. flows and temperature.

In addition, the semiconductor substrate **110** may comprise a first vertical nanowire **125** having a first diameter and a second vertical nanowire **125** having a second diameter different from the first vertical nanowire. In the case of epitaxial growth of the vertical nanowires **125**, the first and the second vertical nanowire may e.g. be formed by using catalyst nanoparticles having different diameters.

Now referring to FIG. 2, the method proceeds by depositing a protection layer **140** on the semiconductor substrate and around a bottom portion **125b** of the vertical nanowire **125**. The protection layer **140** may be formed of one or several organic or inorganic layers, such as Hydrogen silsesquioxane (HSQ), SiNx, SiO₂, BCB, and photoresists. The protection layer **140** may be deposited in a manner which per se is known in the art such as spin coating. A typical thickness of the protection layer **140** may be in a range of 10 to 100 nm. Thereby a typical height of the bottom portion **125b** of the vertical nanowire **125** may be in a range of 10 to 100 nm.

Now referring to FIG. 3, the method proceeds by forming a top contact **150** around a remaining portion **120t** of the vertical nanowire **125** not covered by the protection layer **140**. FIG. 3 shows that the top contact **150** is formed on the top portion **125t** of the vertical nanowire **125** not covered by the protection layer **140**. The top contact **150** may be formed of any of metals such as W, TiN, and Ni. The top contact **150** may comprise a layer or a stack of layers. The top contact **150** may be formed in a manner which per se is known in the art such as sputtering and atomic layer deposition (ALD). A typical thickness of the top contact **150** along the X axis or the Z axis may be in a range of 3 to 50 nm. For instance, the top contact **150** may be formed by sputtering of 20 nm W, ALD of 5 nm TiN or 5 nm of Ni.

Now referring to FIG. 4, the method proceeds by removing the protection layer **140**. The step of removing the protection layer **140** may be done by wet etching of the protection layer **140**. The wet etching of the protection layer **140** may be performed using hydrofluoric acid (HF). The step of removing the protection layer **140** from the bottom portion **125b** of vertical nanowire **125** may proceed until the bottom portion **125b** of vertical nanowire **125** is exposed. FIG. 4 shows the vertical nanowire **125** subsequent to the step of removing the protection layer **140** from the bottom portion **125b** of vertical nanowire **125**. FIG. 4 shows that the bottom portion **125b** of vertical nanowire **125** is exposed.

Now referring to FIG. 5, the method proceeds by depositing a spacer layer **160** on the semiconductor substrate **110**. FIG. 5 shows that the spacer layer **160** is deposited around a lower portion **120l** of the bottom portion **125b** of the vertical nanowire **125**. FIG. 5 further shows that a thickness of the spacer layer **160** is smaller than the thickness of the protection layer **140**. FIG. 5 further shows that a remaining portion of the bottom portion **125b** of the vertical nanowire **125**, not covered by the spacer layer **160**, comprises an intermediate portion **125i** and an upper portion **125u**. The spacer layer **160** may be formed of an inorganic material such as SiO₂ or an organic material such as BCB. The spacer layer **160** may be deposited in a manner which per se is known in the art such as plasma enhanced chemical vapor deposition (PECVD), CVD, spin-coating or ALD. A typical thickness of the spacer layer **160** may be in a range of 5 to 150 nm.

Now referring to FIG. 6, the method proceeds by removing a shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. FIG. 6 shows that a core portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** has been exposed subsequent to the step of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. The step of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** may be performed by etching. The etching may be performed in a manner which per se is known in the art such as using cyclic etching using e.g. hydrochloric acid (HCl) and oxygen treatment. The etching may be performed in-situ or by transferring the samples between different equipment. The chemicals may further be selected to either be selective or non-selective with respect to the core and the shell materials.

The upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be masked, prior to the step of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. The masking of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed in a manner which per se is known in the art.

Now referring to FIG. 7a, the method proceeds by trimming a shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. FIG. 7a shows that a thickness of the shell portion at the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** has been reduced. FIG. 7a shows that the shell thickness at the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** is smaller than the shell thickness at the lower portion **120l** of the bottom portion **125b** of the vertical nanowire **125** i.e. the shell thicknesses at the upper portion **125u** and the lower portion **120l** of the bottom portion **125b** of the vertical nanowire **125** are asymmetric. In the case that the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** has been masked, prior to the step of

removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**, the method may further comprise unmasking the upper portion (**125u**) of the bottom portion (**125b**) of the vertical nanowire (**125**), prior to the step of trimming the shell portion of the upper portion (**125u**) of the bottom portion (**125b**) of the vertical nanowire (**125**).

The step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed by etching. The step of trimming may be performed in a manner similar to the step of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. The step of trimming may be controlled such that a desired thickness of the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be removed. The step of trimming may e.g. be controlled by controlling an etching time. The thickness of the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be reduced by a factor of 30-80%, subsequent to the step of trimming. The intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** may be masked, prior to the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. The masking of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** may be performed by repeating the above described process steps with a new mask that has a different thickness as compared to the mask used for masking the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**.

Now referring to FIG. **7b**, the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may comprise removing the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. FIG. **7b** shows that a core portion of the vertical nanowire **125** at the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** has been exposed. In other words, the upper portion **125u** and the lower portion **125l** of the bottom portion **125b** of the vertical nanowire **125** have become asymmetric.

Still referring to FIG. **7b**, the step of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** and the step of removing the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed by etching. The steps of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** and the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed simultaneously. For instance, steps of removing the shell portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** and the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed in a same etching process using the same etchant.

The method may comprise forming a gate oxide layer around the core portion of the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. The gate oxide layer may be formed of at least a high-k dielectric material such as HfO_x . The gate oxide layer may be formed in a manner which per se is known in the art such as ALD. A typical thickness of the gate oxide layer may be in a range of 1 to 6 nm. An example of gate oxide layer may be $\text{Al}_2\text{O}_3/\text{HfO}_2$ with a typical thickness of 1 nm Al_2O_3 .

Now referring to FIGS. **8a** and **8b**, the method proceeds by depositing a metal gate **170** on the spacer layer **160** and around the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** i.e. depositing the metal gate **170** on the spacer layer **160** and around the gate oxide layer if present. The metal gate **170** may be deposited in a manner which per se is known in the art such as sputtering and ALD. The metal gate **170** may be formed of at least a metal such as TiN and W. A typical thickness of the metal gate **170** may be in a range of 5 to 100 nm.

Referring to FIG. **8a**, the step of depositing the metal gate **170** on the spacer layer **160** and around the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** may be performed prior to the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. In other words, the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125** may be performed subsequent to the step of depositing the gate oxide layer and the metal gate **170** on the spacer layer **160** and around the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125**. In this way, the intermediate portion **125i** of the bottom portion **125b** of the vertical nanowire **125** may be protected by the metal gate **170** during the step of trimming the shell portion of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**.

Now referring to FIGS. **9a** and **9b**, the method may proceed by depositing another spacer layer **180** e.g. a second spacer layer **180** on the metal gate **170** and around at least the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. The second spacer layer **180** may be deposited in a similar manner as of the spacer layer **160** i.e. the first spacer layer **160**. A thickness of the second spacer layer **180** may be in a range of 20 to 200 nm. The second spacer layer **180** may be deposited such that the thickness of the second spacer layer **180** may be similar to a height of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**. The second spacer layer **180** may be deposited such that the thickness of the second spacer layer **180** may be larger than the height of the upper portion **125u** of the bottom portion **125b** of the vertical nanowire **125**.

Now referring to FIGS. **10a** and **10b**, the method proceeds by forming a lower and an upper source drain portions **110** and **190**. The lower and the upper source drain portions may respectively be comprised in a lower and an upper source drain contacts. FIGS. **10a** and **10b** show an example of an upper source drain portion **190** i.e. a metal upper source drain portion **190** has been deposited on the second spacer layer **180** and around the top portion **125t** of the vertical nanowire **125**. The upper source drain portion **190** may have a thickness similar to a height of the top portion **125t** of the bottom portion **125b** of the vertical nanowire **125**. The upper source drain portion **190** may have a thickness different than the height of the top portion **125t** of the bottom portion **125b** of the vertical nanowire **125**. The upper source drain portion **190** may be deposited in a manner which per se is known in the art such as sputtering.

In the case that the semiconductor substrate **110** comprises a highly doped portion e.g. a highly doped InAs layer formed on the Si substrate, that highly doped portion may be formed as the lower source drain portion. As shown in FIGS. **10a** and **10b**, the semiconductor substrate **110** acts as the lower source drain portion **110**. The lower source drain portion may be formed in a manner which per se is known

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in the art. For instance, the lower source drain portion may be formed by patterning of the highly doped portion using e.g. electron beam lithography.

Still referring to FIGS. 10*a* and 10*b*, these figures schematically illustrate two examples of the asymmetric vertical nanowire MOSFETs 100. FIGS. 10*a* and 10*b* show two asymmetric vertical nanowire MOSFETs 100 comprising at least one vertical nanowire 125 arranged on a semiconductor substrate 110. FIGS. 10*a* and 10*b* further show that the at least one vertical nanowire 125 comprises a core portion 120 and a shell portion 130. FIGS. 10*a* and 10*b* show that the shell portion 130 circumscribes the core portion 120 along a height of the vertical nanowire 125. FIGS. 10*a* and 10*b* further show a first spacer layer 160 arranged on the semiconductor substrate 110 and around a lower portion 125*l* of a bottom portion 125*b* of the vertical nanowire 125. FIGS. 10*a* and 10*b* further show a metal gate 170 arranged on the first spacer layer 160 and around an intermediate portion 125*i* of the bottom portion 125*b* of the vertical nanowire 125. FIGS. 10*a* and 10*b* show that the intermediate portion 125*i* of the bottom portion 125*b* of the vertical nanowire 125 comprises a through-going recess in the shell portion such that the recess circumscribes the core portion 120.

FIGS. 10*a* and 10*b* show that a second spacer layer 180 is arranged on the metal gate 170 and at least around an upper portion 125*u* of the bottom portion 125*b* of the vertical nanowire 125. FIGS. 10*a* and 10*b* further show that the shell portion at the upper portion 125*u* of the bottom portion 125*b* of the vertical nanowire 125 has a reduced thickness compared to the lower portion 125*l* of the bottom portion 125*b* of the vertical nanowire 125. FIGS. 10*a* and 10*b* further show a top contact 150 arranged around a top portion 125*t* of the vertical nanowire 125. FIGS. 10*a* and 10*b* show that the shell portion of the top portion 125*t* of the vertical nanowire 125 has a corresponding thickness compared to the lower portion 125*l* of the bottom portion 125*b* of the vertical nanowire 125. FIGS. 10*a* and 10*b* further show a lower source drain portion 110 and an upper source drain portion 190. The asymmetric vertical nanowire MOSFETs 100 presented in FIGS. 10*a* and 10*b* demonstrate transistors with an asymmetric shell structure in the direct vicinity of the gate metal 170. The asymmetric shell structure provides, on one side of the metal gate, a low access resistance provided by the presence of the shell portion 130. The asymmetric shell structure provides, on the other side of the metal gate, a low capacitance provided by the separation of the shell portion 130 and the metal gate 170. Combined these two properties provide advantages in the design of the high-frequency transistors and including their gain properties.

In the above the inventive concept has mainly been described with reference to a limited number of examples. However, as is readily appreciated by a person skilled in the art, other examples than the ones disclosed above are equally possible within the scope of the inventive concept, as defined by the appended claims. The person skilled in the art realizes that the present invention by no means is limited to the preferred embodiments described above. On the contrary, many modifications and variations are possible within the scope of the appended claims.

For example, a portion of the core portion of the intermediate portion of the bottom portion of the vertical nanowire may be removed, subsequent to removing a shell portion of the intermediate portion of the bottom portion of the vertical nanowire. Another example, a portion of the core portion of the upper portion of the bottom portion of the vertical nanowire may be removed, subsequent to removing a shell portion of the upper portion of the bottom portion of

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the vertical nanowire. Additionally, variations to the disclosed embodiments can be understood and effected by the skilled person in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims.

What is claimed is:

1. A method for fabricating an asymmetric vertical nanowire MOSFET on a semiconductor substrate comprising at least one vertical nanowire, wherein the at least one vertical nanowire comprises a core portion and a shell portion circumscribing the core portion along a height of the vertical nanowire, the method comprising:

depositing a protection layer on the semiconductor substrate and around a bottom portion of the vertical nanowire,

forming a top contact around a remaining portion of the vertical nanowire not covered by the protection layer,

removing, by etching, the protection layer,

depositing a spacer layer on the semiconductor substrate and around a lower portion of the bottom portion of the vertical nanowire, wherein a thickness of the spacer layer is smaller than the thickness of the protection layer and wherein a remaining portion of the bottom portion of the vertical nanowire, not covered by the spacer layer, comprises an intermediate portion and an upper portion,

removing a shell portion of the intermediate portion of the bottom portion of the vertical nanowire, thereby exposing a core portion of the vertical nanowire at the intermediate portion of the bottom portion of the vertical nanowire,

masking the upper portion of the bottom portion of the vertical nanowire, prior to the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire,

trimming a shell portion of the upper portion of the bottom portion of the vertical nanowire, thereby reducing a thickness of the shell portion of the upper portion of the bottom portion of the vertical nanowire or removing the shell portion of the upper portion of the bottom portion of the vertical nanowire,

depositing a metal gate on the spacer layer and around the intermediate portion of the bottom portion of the vertical nanowire, and

forming a lower and an upper source drain portions.

2. The method according to claim 1, wherein the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire comprises removing the shell portion of the upper portion of the bottom portion of the vertical nanowire, thereby exposing a core portion of the vertical nanowire at the upper portion of the bottom portion of the vertical nanowire.

3. The method according to claim 1, wherein the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire are performed by etching.

4. The method according to claim 1, wherein the step of removing the shell portion of the intermediate portion of the bottom portion of the vertical nanowire and the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire are performed simultaneously.

5. The method according to claim 1, wherein the step of depositing the metal gate on the spacer layer and around the intermediate portion of the bottom portion of the vertical

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nanowire is performed prior to the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire.

6. The method according to claim 5, further comprising: unmasking the upper portion of the bottom portion of the vertical nanowire, prior to the step of trimming the shell portion of the upper portion of the bottom portion of the vertical nanowire.

7. The method according to claim 5, wherein the core portion of the vertical nanowire is formed of InAs, InGaAs, or a combination thereof.

8. The method according to claim 1, wherein the shell portion of the vertical nanowire is formed of InGaAs with an n-type dopant dose of at least 1.10^{16} atoms/cm³.

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